$$\mathrm{Tr}_{i,j}^{k}\, f = \mathrm{Tr}_{i,j}^{k}\, \mathbf{A} = \mathbf{A}_{i,j} + \mathbf{A}_{k,j}(\mathbf{1} - \mathbf{A}_{k,k})^{-1}\mathbf{A}_{i,k}$$